23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I & PHASE – II

Analysis and Architectural Modelling (CO1 & CO2)

Group No : ______B1____________

Team Members :

Roll No	Student Name
AM.SC.U4CSE25141	Parvathy S Lal
AM.SC.U4CSE25138	Neha B Nair
AM.SC.U4CSE25117	Ganne Neha Abhigna
AM.SC.U4CSE25166	Janwi Yadhav

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

1. Problem Title: Personal Task Manager System

2. Problem Description: A personal task manager helps users organize daily activities, track pending tasks, and manage deadlines. The system should allow users to add tasks, update task status, and view task lists.

3. Identified Entities (Classes)

Class Name	Description
User	Stores user details and interacts with the system.
Task	Stores task information like name, deadline, and status.
TaskManager	Manages tasks such as adding, updating, deleting, and displaying.
Category	Classifies tasks into different groups.
FileManager	Handles saving and loading tasks from files.

4. Class Attributes and Behaviours

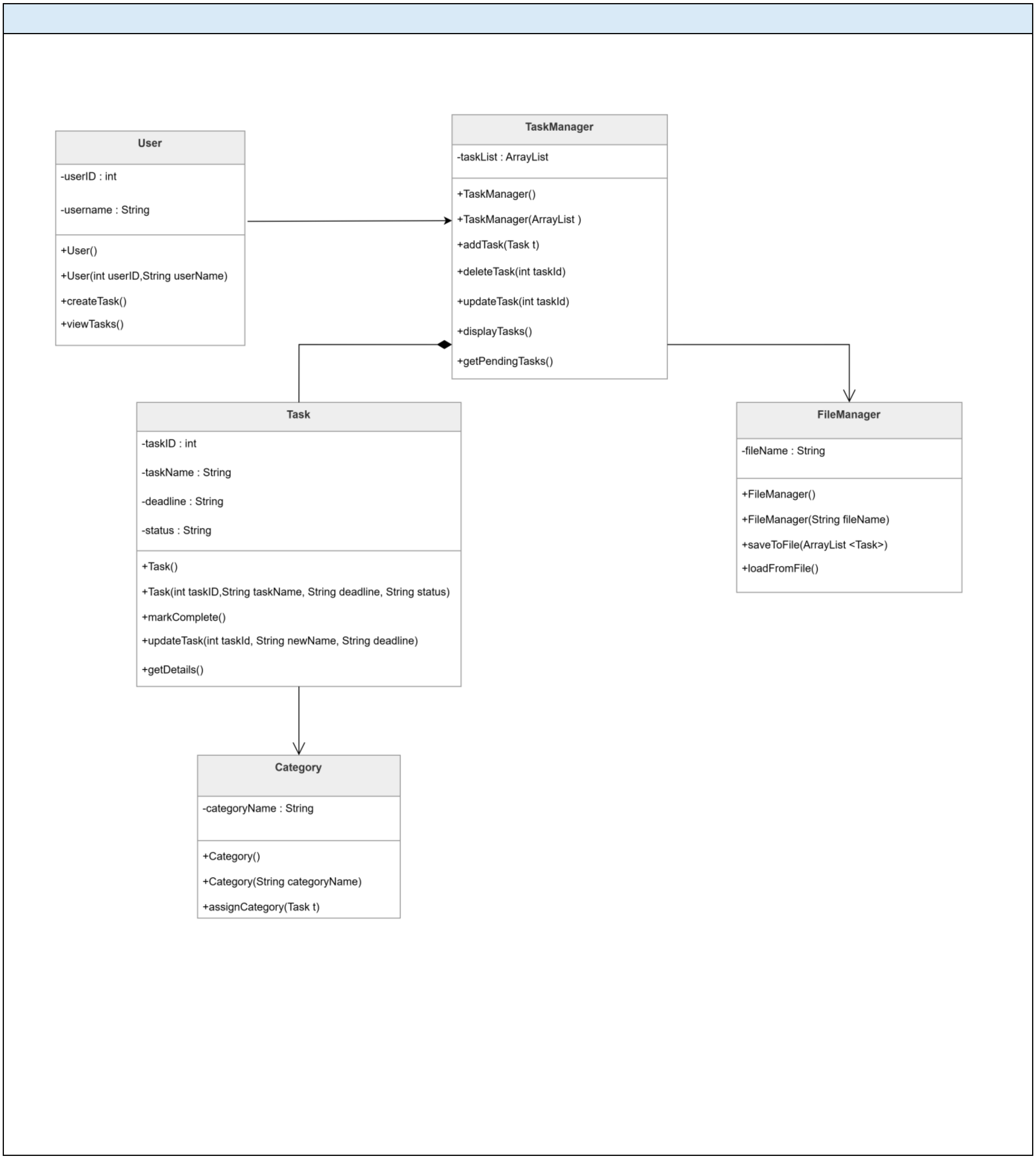
Class Name	Attributes	Methods (Behaviours)
User	userID username	createTask() viewTasks()
Task	taskID taskName deadline status	markComplete() updateTask(int taskId, String newName, String deadline) getDetails()
TaskManager	taskList	addTask(Task t) deleteTask(int taskId) updateTask(int taskId) displayTasks() getPendingTasks()
Category	categoryName	assignCategory(Task t)
FileManager	fileName	saveToFile(ArrayList<Task>) loadFromFile()

PHASE – II (Architectural Modelling)

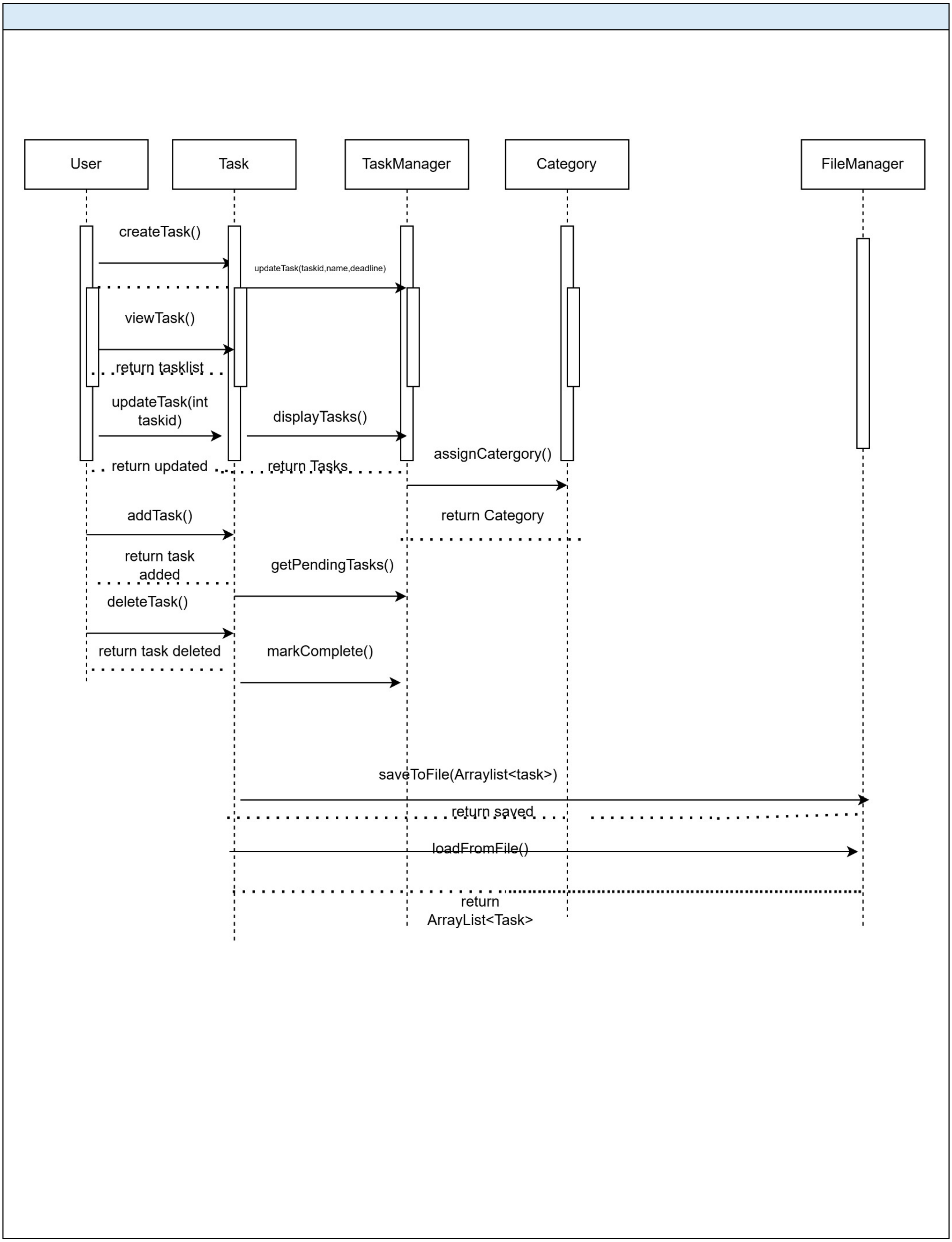
UML Class Diagram and Sequence Diagram (CO2)

1. Problem Title: Personal Task Manager System

2. UML Class Diagram



3. Sequence Diagram



4. Brief Explanation of the Design

- **Major classes identified and their responsibilities:**
  The main classes in the system are User, Task, TaskManager, Category, and FileManager. The User interacts with the system to create and view tasks. The Task class stores task details such as task ID, task name, deadline, and status, and provides operations like marking a task as complete and updating task details. The TaskManager is the central class that manages the collection of tasks by adding, deleting, updating, and displaying them. The Category class is used to group or classify tasks under different categories. The FileManager is responsible for saving task data to a file and loading it back when needed.

- **Relationships used (association / aggregation / inheritance, if any):**
  The design mainly uses association and composition style relationships. There is an association between User and TaskManager, because the user interacts with the task manager to perform operations. There is a strong has-a relationship between TaskManager and Task, since the task manager maintains the task list. There is also an association between Task and Category, because each task can be assigned to a category. The TaskManager is also associated with FileManager for saving and loading task details.

- **Key interaction flow captured in the sequence diagram:**
  The sequence diagram shows how the User sends a request to the TaskManager to perform an action such as adding, updating, deleting, or viewing tasks. The TaskManager then interacts with the Task objects to make the required changes. If categorization is needed, the task may also be linked with the Category class. When task data needs to be stored permanently or retrieved later, the TaskManager communicates with the FileManager to save or load the task list. Thus, the sequence diagram captures the flow of control from the user to the manager class and then to the supporting classes.